

CIA 3 : ML FOR SOCIAL GOOD CHALLENGE

Title : Heart Disease Prediction using Ensemble Machine Learning

Mission : Health

Problem

Predict whether a person has a high risk of heart disease based on clinical attributes to support early risk screening and timely medical evaluation.

Dataset

The official UCI repository describes it as a health/medicine classification dataset with 303 instances and 13 features. The target represents the presence/absence of heart disease, and the commonly used formulation converts the target into 0 = absence and 1 = presence.

https://archive.ics.uci.edu/dataset/45/heart+disease?utm_source=chatgpt.com

1. Social problem

Heart disease is a major non-communicable disease, and identifying individuals who may be at higher risk can support earlier medical evaluation.

2. Intended beneficiaries

Our intended beneficiaries are:

- Individuals who may be at risk
- Healthcare professionals
- Primary healthcare/under-resourced clinics

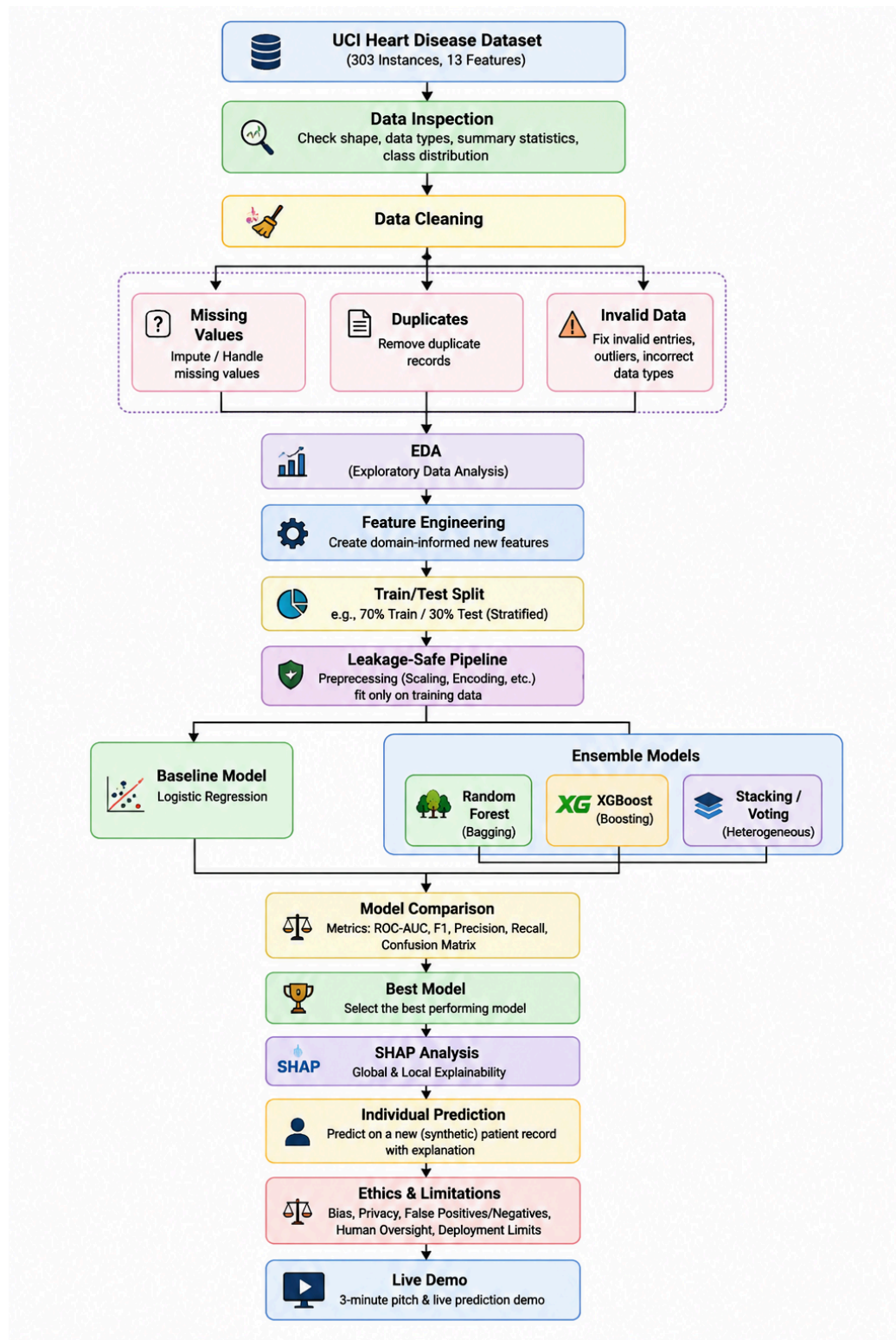
3. Prediction target

Our target will be:

0 → No presence of heart disease

1 → Presence of heart disease

The UCI documentation explains that the original target has values 0–4 and that experiments commonly convert it into absence versus presence.



Purpose

This project was developed for educational purposes to demonstrate the application of machine learning and ensemble learning techniques for heart disease prediction. The model is intended to support learning and experimentation and should not be used as a substitute for professional medical diagnosis.

Responsible AI Principles

This project follows responsible AI practices by emphasizing transparency, fairness, and ethical use of predictive models.

- Predictions should support—not replace—clinical decision-making.
- Patient privacy and confidential health information must be protected.
- Machine learning models should be evaluated for fairness and potential bias before real-world deployment.
- Healthcare professionals should validate all predictions before making treatment decisions.

Limitations

The model is trained on the UCI Cleveland Heart Disease Dataset and may not generalize to all patient populations, hospitals, or clinical environments. The results should therefore be interpreted with appropriate caution.

Explainability

To improve transparency, the project uses SHAP (SHapley Additive exPlanations) to explain both global feature importance and individual predictions, helping users understand how different clinical features influence the model's decisions.

Commitment

This project promotes the responsible use of artificial intelligence by encouraging human oversight, transparent model explanations, ethical handling of health-related data, and the educational use of predictive analytics in healthcare.